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REVIEW

Anterior Cruciate Ligament Tears among Football Players

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Abstract

Purpose of Review Anterior Cruciate Ligament (ACL) tears are one of the most common causes of lost playing time in American football athletes. Recently, there has been a push to get athletes back to sport faster. As such, numerous studies have evaluated management and rehabilitation protocols for return to play after ACL injury in football players. The purpose of this review is to synthesize information, both classic and new, to aid orthopedic surgeons in treatment of football players with ACL injuries.

Recent Findings Recent studies have demonstrated that not all ACL injuries are alike. Management should be a shared decision-making process between athlete and surgeon. Studies have demonstrated low failure rates when using bone-patellar tendon-bone (BTPB) autograft which is the most common graft utilized for elite football athletes. Outcomes are continually being evaluated by multicenter study groups such as the Multicenter Orthopaedic Outcome Network which has established a thorough rehabilitation protocol focusing on athlete milestones.

Summary ACL tears in the football athlete are common and challenging injuries. Treatment revolves around ACL reconstruction, most commonly with BTPB autograft. Post-operative rehabilitation is essential and should focus on objective criteria rather than time elapsed. Return to play criteria rely upon symptoms, athlete confidence, strength, and both functional and football specific testing. Return-to-play timelines are individualized for each football athlete but recent trends have highlighted a more delayed return of at least 7-9 months. Rate of returning to play varies from 63-82% and depends upon many factors including level of play and position.

Keywords ACL tear · Football injury · Operative · Non-operative · Return to sport

Introduction

The anterior cruciate ligament (ACL) has long been known to be a critical stabilizer of the knee joint. Galen described the ACL as a primary component to prevent abnormal knee motion as early as 165 A.D [1]. Since that time, countless articles have been published about ACL tears and the medical field has collectively improved in our ability to diagnose and treat these injuries. Unfortunately, ACL tears continue to be a problem for athletes and non-athletes alike. The purpose of this review is to report on the ever-changing landscape of ACL injury prevention, treatment, rehabilitation, and outcomes in American football players.

Anterior cruciate ligament (ACL) tears are a common cause of lost playing time in American football athletes. This injury unfortunately impacts athletes of all ages and levels of play and has been studied extensively in high school athletes up through professional football players. Patients generally present acutely after their injury with knee pain, a palpable effusion, and decreased range of motion. Radiographs and MRI aid in the confirmation of the diagnosis and to assist with treatment strategies. Treatment is individualized, but in 2024 the mainstay is arthroscopic-assisted ACL reconstruction. Postoperative rehabilitation protocols are being studied

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to optimize recovery and aid in rates and timing of return to play while limiting re-injury rates. Questions remain regarding graft selection, rehabilitation protocols, and timing of return to sport. The purpose of this review is to synthesize classic and new data to aid surgeons in managing these common but difficult injuries.

Epidemiology

Numerous studies have evaluated the incidence of ACL tears in football athletes of varying levels of play. Joseph et al. performed a study utilizing online injury forms submitted by athletic trainers at 100 high schools throughout the United States. They found that the approximate incidence for high school football players was about 11.1 per 100,000 athlete exposures. In their high school athlete population, the incidence of ACL tears among male football players was second only to girls' soccer [2].

Unfortunately, the risk of ACL injury remains high in collegiate and professional football players. National Collegiate Athletic Association (NCAA) football players have been reported to have ACL injuries at an incidence rate of 14.4–18.0 per 100,000 athlete exposures [3]. At the highest levels of play in the National Football League (NFL), the single-season ACL injury risk has been reported to be 1.9% [4].

High school athletes often face increased injury risk due to factors like improper training techniques, insufficient conditioning, and immature neuromuscular control [5]. In college football, intensity and competitiveness increases and contribute to higher rates of ACL tears compared to high school. Professional football players, while benefiting from excellent training and medical resources, still encounter ACL tear risks due to the demands of the sport and increased contact forces. Understanding these epidemiological patterns is crucial for implementing preventive strategies and optimizing rehabilitation protocols to mitigate ACL injury risk in football players across all levels of competition.

Mechanism of Injury

ACL tears represent a spectrum of injuries with various etiologies, categorized broadly into contact and non-contact mechanisms of injury. Contact injuries occur when direct force is applied to the knee, often during collisions or tackles. In these scenarios, the ACL may tear due to excessive external force, leading to sudden hyperextension, valgus, or rotational stress on the knee.

Conversely, non-contact ACL tears commonly arise during activities involving rapid deceleration, cutting, or pivoting motions [6, 7]. These injuries often result from dynamic movements where the lower limb experiences an abrupt change in direction, causing the femur to internally rotate relative to the fixed tibia, thereby placing undue strain on the ACL until it ruptures [8] (Fig. 1).

In a recent study of ACL tears in the NFL by Brophy et al., video footage of 140 ACL tears over a 6-year span were examined and 30% occurred via direct contact to the injured lower extremity. However, this varied by position. For example, ACL tears in offensive lineman occurred by direct contact in just over two-thirds of the injuries to the lower extremity while wide receivers did not have any direct contact ACL tears [9].

In addition to rupture of the ACL, football athletes commonly have concomitant injuries like collateral ligament injuries, meniscal injuries, or bony contusions, among others (Fig. 2). Brophy et al. examined MRIs of NFL athletes following ACL tears and found meniscal, ligamentous, or cartilage injury in 89% of knees. Meniscal tears were found in 70% of the MRIs, 59% involving the lateral side and 41% medial. Concurrent ligamentous injury occurred in 71% of MRIs, most commonly involving the MCL (57%). MCL injuries, when present, were grade 1 or 2 in 67% of knees and grade 3 in 33%. Bone bruises were demonstrated in 95% of ACL tears, most commonly (81%) involving the lateral tibial plateau. Chondral pathology was found in 49% of all MRIs, but only 25% of these were full thickness [10].

Fig. 1 Sagittal slice of a left knee T2 MRI of a 20-year-old American football athlete with an acute ACL tear

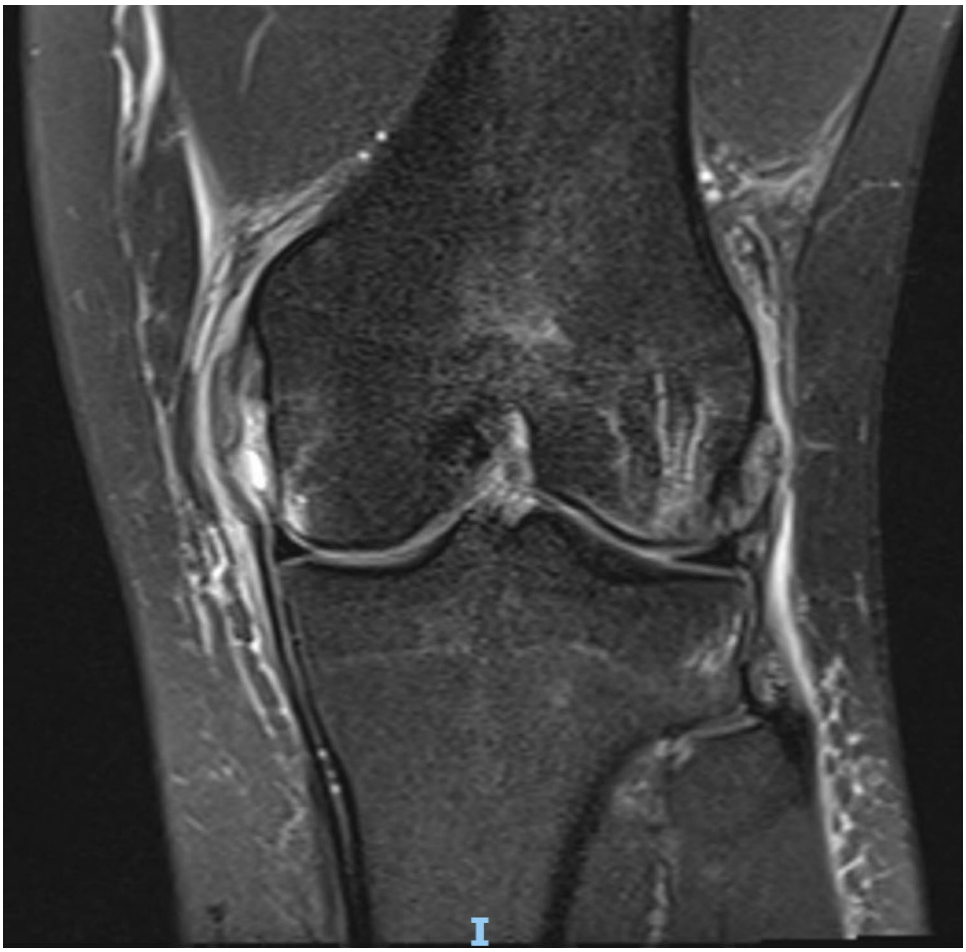

Fig. 2 Coronal slice of a left knee T2 MRI of a 20-year-old American football player with an acute ACL tear. There are associated contusions of the lateral femoral condyle and posterior medial tibial plateau. In addition, this athlete sustained a high-grade proximal medial collateral ligament injury

Treatment Options

Although there is a relative high incidence of ACL injuries within football athletes, there is still debate amongst orthopedic surgeons regarding the ideal management. Discussions of non-operative management may be a part of the ACL tear treatment algorithm. This would include a strict rehabilitation protocol with physical therapy to strengthen the muscles around the knee, notably the quadriceps femoris and hamstrings [11]. However, given the risk for recurrent instability, secondary soft tissue pathology including cartilage damage and meniscal tears, and possible long-term risk of osteoarthritis, the discussion of treatment for these young and active patients who desire to return to cutting and pivoting activities centers around surgical intervention [12].

Options regarding surgical intervention include both repair and reconstruction of the ACL. Although repair is a possibility, especially in the setting of newer techniques such as the Bridge-enhanced ACL repair (BEAR), there is a paucity of literature demonstrating superiority over reconstruction [13, 14]. The gold standard for surgical management of ACL tears remains reconstruction. Overall, the management of ACL tears should be a discussion between provider and athlete and include many factors including patient-specific characteristics,

such as comorbidities, skeletal maturity, and importantly future desired activity including both return to sport expectations and future work.

The ideal graft choice for ACL reconstruction remains controversial. Currently, the most common graft options include bone-patellar tendon-bone (BPTB), quadriceps tendon (QT), hamstring tendon (HS), and allograft. Over time, BPTB autograft has been considered the gold standard due to bone-to-bone healing, ease of harvest, and its biomechanical properties [15]. Autograft BPTB has also been demonstrated to be superior to autograft HS in a large series of high school and college athletes with HS graft requiring revision 2.1 times more frequently [16]. However, more recently all soft-tissue autografts including QT and HS have been utilized and commonly result in less kneeling pain with comparable outcomes. Due to the increased risk of graft failure, and meniscus repair failure, the usage of allograft for ACL reconstruction in competitive football players has gone down significantly. [17] The choice of graft is very frequently a combination of surgeon preference and patient factors. [17–19] In a 2008 survey, Schrock et al. demonstrated

that BPTB autografts were the gold standard within NFL team physicians: 87% of their ACL reconstructions being completed with BPTB autograft. In 2016, a repeat survey was completed and the percent utilizing BPTB autografts increased to 97%. [20] This may be due to the concern that all soft-tissue grafts incorporate more slowly into bone tunnels when compared to BPTB autografts. [21] It is the senior author's preference to use BPTB autograft for skeletally mature competitive high school and college football players, and to use QT autograft in skeletally immature football athletes.

The majority of ACL research has demonstrated that the risk for ACL re-injury is younger age and higher activity level. Additional procedures are now being occasionally added in addition to ACL reconstruction for football players. Players with risk factors for recurrent injury have been identified. This includes significant (≥ 5–10º) knee hyperextension, and high-grade Lachman or pivot shift examination. [22] Usage of procedures such as lateral extra-articular tenodesis (LET) or anterolateral ligament (ALL) reconstruction have been shown to lower risk of HS graft rupture when added to ACL reconstruction. [23, 24] There continues to be debate regarding the indications for adding LET or ALL procedures but there is a growing trend to consider its usage in competitive football athletes. It is the senior author's preference to add LET to ACL reconstruction in patients with knee hyperextension, significant hyperlaxity as graded with Beighton Scale, grade III laxity on Lachman, an explosive pivot shift examination, and in revision ACL reconstruction situations.

Rehabilitation Protocols

Whether pursuing non-operative or operative intervention for an ACL tear, rehabilitation is essential for re-establishing knee range of motion, lower extremity strength, power, and neuromuscular control. Overall recommendations are similar in both non-operative and operative groups and are dependent upon progression of milestones over set timelines. [25] Rehabilitation phases should be progressed based on objective criteria, many of which have been established within the Multicenter Orthopaedic Outcome Network (MOON) ACL Rehabilitation Guidelines [26].

In the early phase of rehabilitation, the goals revolve around reducing pain and swelling, regaining range of motion, achieving good quadriceps activation, maintaining lower extremity strength, and normalizing gait. Utilization of neuromuscular electrical stimulation to the quadriceps as well as blood flow restriction therapy during the early phase of rehabilitation may help the athlete to more quickly overcome arthrogenic quadriceps muscle inhibition and improve strength. [27–29] Once pain and swelling have reduced, progression to more traditional strength training including both open and closed chain exercises may commence with a primary focus on regaining quadriceps strength. Additionally, posterior chain strengthening and core and hip neuromuscular control exercises are included during this phase to improve stability and dynamic lower extremity control.

Once the athlete demonstrates sufficient quadriceps strength and function, ideally tested via handheld dynamometry or isokinetic device and functional testing, activities such as plyometrics and linear accelerations and decelerations are initiated. Intensities of these activities are gradually progressed based on the athlete's abilities and confidence. During this intermediate phase of rehabilitation typically around 3–4 months post-operative, focus remains on improving quadriceps strength but also on developing power and the ability for the muscle to decelerate forces quickly.

Late-stage rehabilitation includes the addition of sport-specific training including on-the-field activities. For the football athlete, these include change of direction activities beginning with submaximal efforts in controlled environments with gradual progressions to higher intensities and reacting to environmental stimuli. This type of training is often initiated around 6 months post-operative but demonstration of readiness via passing return-to-sport testing criteria including psychological readiness, isokinetic quadriceps and hamstring strength, and functional testing is critical. [30–32] Coordination of care and shared decision-making between the athlete, athletic trainer, physical therapist, physician, sport psychologist, and strength and conditioning staff is important during this phase to ensure the athlete is safely integrating back into sport-specific training and eventually competition level performance, likely closer to 9 months postoperative [33, 34].

Return to Play and Outcomes

Regarding return to play timelines, orthopedic surgeons typically do not allow RTP until 6 to 12 months post-op. Schrock et al. found in 2008 that 49% of NFL team physicians stated they would allow RTP 6 months after ACL. This survey was revisited in 2016 and only 14% of team physicians would allow RTP at 6 months with most delaying until 7 to 9 months post-operatively. [20] Erickson et al. published a study in 2014 that found only 56.2% of NFL and NCAA team physicians would allow RTP at 6 months and 12.4% would not allow RTP in less than 9 months. [35] Ross et al. in 2020 preformed a systematic review of the current literature on NFL and NCAA football athletes returning to sport and found that the weighted mean time to RTP after ACL reconstruction was 11.6 months [36].

McCullough et al. in the Multicenter Orthopaedic Outcomes Network (MOON) cohort demonstrated that 63% of

high school football players and 69% of collegiate football players returned to play. [37] A study by Darulwalla et al. evaluated the rate of RTP following ACL reconstruction among NCAA Division 1 collegiate football athletes and found that 82% of the athletes, including 94% of starters, were able to RTP. Factors associated with higher rates of RTP included higher placement on the depth chart and those on athletic scholarship. They found no significant difference between players who underwent an isolated ACL reconstruction versus those who underwent additional procedures [38].

Literature evaluating RTP rates of NFL athletes have found rates varying from 63 to 73%. [36, 39, 40] Of note, running backs, wide receivers, and tight ends had lower rates of return to play, while defensive players, quarterbacks, and kickers had higher rates of RTP. [35, 39, 41, 42] These studies demonstrated the difficulty in RTP for skill position players who rely more heavily on cutting and pivoting movement.

Dai et al. conducted a systematic review and meta-analysis of studies not specific to athletes and demonstrated a graft failure rate of 4.1% in QT autografts and 2.0% of BPTB autografts, $p = 0.36$. However, QT and BPTB autografts were associated with similar IKDC subjective scores. When comparing graft failure rates between QT and HS autografts, there was also a higher rate of graft failures in the QT autograft group, 3.7% versus 2.5%, respectively with $p = 0.57$. QT and HS autografts were associated with similar IKDC subjective scores. [43] Similarly, a recent meta-analysis by Samuelson et al. comparing BPTB autografts to HS autografts in over 47,000 patients demonstrated graft failure in 2.84% of HS autografts versus 2.80% in BPTB autografts [44].

Lai et al. performed a meta-analysis of the ACL graft failure specific to elite athletes including football players and demonstrated an overall higher rate of 5.2%. [45] Also specific to a younger, athletic population, Spindler et al. of the MOON group demonstrated the incidence of subsequent ACL reconstruction at 6 years was 9.2% in the ipsilateral knee and 11.2% in the contralateral normal knee. Additionally, the incidence of ACL graft revisions at 6 years following index surgery was 2.1 times higher with HS autograft when compared to BPTB autograft. [16] The MOON group also evaluated the 10-year incidence of radiographic post-traumatic osteoarthritis after ACL reconstruction and found 37% of knees to have osteophytes and 23% have joint space narrowing [46].

Conclusion

ACL tears in the football athlete are common and challenging injuries. Treatment revolves around ACL reconstruction, most commonly with BTPB autograft. Post-operative rehabilitation is essential and should focus on objective criteria rather than time elapsed. Return to play criteria rely upon symptoms, athlete confidence, strength, and both functional and football specific testing. Return to play timelines are individualized for each football athlete but recent trends have highlighted a more delayed return of at least 7–9 months. Rate of returning to play varies from 63–82% and depends upon many factors including level of play and position.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

This study did not receive any financial support.
The figures do not require publication in color.
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Competing interests The authors declare no competing interests.

Disclaimer None

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